

Lab-7

Updated Model.py

```
from django.db import models
from django.contrib.auth.models import User

class Course(models.Model):
    title = models.CharField(max_length=25)
    descriptions = models.TextField()
    durations = models.IntegerField(help_text="Durations in hours")
    thumbnail = models.ImageField(upload_to="courses_thumbnail/", null=True, blank=True)
    created_at = models.DateTimeField(auto_now_add=True, null=True, blank=True)

    def __str__(self):
        return self.title

class Lesson(models.Model):
    title = models.CharField(max_length=60)
    content = models.TextField()
    course = models.ForeignKey(Course, on_delete=models.CASCADE, related_name="lessons") # ✓
    video_URL = models.URLField(null=True, blank=True)
    completion_status = models.BooleanField(default=False)

    def __str__(self):
        return self.title

class Student(models.Model):
    user = models.OneToOneField(User, on_delete=models.CASCADE, null=True, blank=True)
    name = models.CharField(max_length=30)
    roll = models.IntegerField()
    email = models.EmailField(max_length=60, unique=True)
    enrolled_courses = models.ManyToManyField(Course, related_name="students") # ✓ Changed
    completed_lesson = models.ManyToManyField(Lesson, related_name='completed_by', blank=True)

    def __str__(self):
        return self.name
```

ORM queries and results in the Django shell.

```
(.venv) abduhhnazmus-sakib@Abdullahs-MacBook-Pro: Edu_Learn_Django_CURD_Project % python3 manage.py shell
9 objects imported automatically (use -v 2 for details).

Python 3.13.0 (v3.13.0:60403a5409f, Oct 7 2024, 00:37:40) [Clang 15.0.0 (clang-1500.3.9.4)] on darwin
Type "help", "copyright", "credits" or "license" for more information.
(InteractiveConsole)
>>> from Course_app.models import Course, Lesson, Student
>>> Course.objects.all()
<QuerySet [<Course: C ++>, <Course: Python>, <Course: JavaScript>, <Course: Bio_Informatics>]>
```

```
>>> course = Course.objects.get(id=2)
>>> course.lessons.all()
<QuerySet [<Lesson: What is C++?>, <Lesson: Why Use C++>, <Lesson: Difference between C and C++>, <Lesson: C++ Syntax>]>
>>> student = Student.objects.get(email="ahmedBulbul@gmail.com")
>>> course = Course.objects.get(id=2)
>>> student.enrolled_courses.add(course)
>>> lesson = Lesson.objects.get(id=14)
>>> student.completed_lesson.add(lesson)
>>> student.completed_lesson.all()
<QuerySet [<Lesson: What is C++?>]>
>>> total_lesson = course.lessons.count()
>>> total_lesson = course.lessons.count()
>>> completed = student.completed_lesson.filter(course=course).count()
>>> progress = (completed / total_lessons) * 100
Traceback (most recent call last):
  File "<console>", line 1, in <module>
NameError: name 'total_lessons' is not defined. Did you mean: 'total_lesson'?
>>> progress = (completed / total_lesson) * 100
>>> print(f"Progress: {progress}%")
Progress: 25.0%
>>>
```

course detail page showing lesson progress.

C++

Duration: 60 hours

Description: C++ is a popular programming language. C++ is used to create computer programmes and is one of the most used languages in game development. C++ was developed as an extension of C, and both languages have almost the same syntax.

Progress: 3/4 lessons completed

75%

Lessons [+ Add New Lesson](#)

Lesson Title	Content	Status	Actions
What is C++?	C++ is a cross-platform language that can be used to create high-performance applications. C++ was developed by Bjarne Stroustrup, as an extension to the C language. C++ gives programmers a high level of control over system resources and memory. The language was updated 5 major times in 2011, 2014, 2017, 2020, and 2023 to C++11, C++14, C++17, C++20, and C++23.	Completed	Edit Delete
	C++ is one of the world's most popular programming languages. C++ can be found in today's operating systems, Graphical User Interfaces, and embedded systems. C++ is an object-oriented programming language which gives a clear structure to programs and allows code		

Updated course_detail.html

```
{% extends 'Course_app/base.html' %}

{% block title %}Course Details{% endblock title %}

{% block content %}
<div class="container">

    <!-- Course Info -->
    <div class="card shadow-lg p-4 mb-4">
        <h2 class="text-primary fw-bold">{{ course.title }}</h2>
        <p class="text-muted mb-1">
            <strong>Duration:</strong>
            <span class="badge bg-success">{{ course.durations }} hours</span>
        </p>
        <p><strong>Description:</strong> {{ course.descriptions }}</p>
    </div>

    <!-- Lesson Progress -->
    <div class="mb-4">
        <h5>Progress: {{ completed_lessons }}/{{ total_lessons }} lessons completed</h5>
        <div class="progress">
            <div class="progress-bar bg-info" role="progressbar"
                style="width: {{ percent_complete }}%"
                aria-valuenow="{{ completed_lessons }}" aria-valuemin="0" aria-valuemax="{{ total_lessons }}">
                {{ percent_complete }}%
            </div>
        </div>
    </div>

    <!-- Add Lesson Button -->
    <div class="d-flex justify-content-between align-items-center my-3">
        <h4 class="fw-bold text-secondary">Lessons</h4>
        <a href="{% url 'create_lesson' %}" class="btn btn-success">
            <i class="fas fa-plus"></i> Add New Lesson
        </a>
    </div>
</div>
```

```

<!-- Lessons Table -->
<div class="table-responsive">
  <table class="table table-bordered table-hover text-center align-middle">
    <thead class="table-dark">
      <tr>
        <th>Lesson Title</th>
        <th>Content</th>
        <th>Status</th>
        <th>Actions</th>
      </tr>
    </thead>
    <tbody>
      {% for lesson in lessons %}
      <tr>
        <td class="fw-bold text-primary">{{ lesson.title }}</td>
        <td class="text-start">
          <div class="border rounded p-2 bg-light" style="white-space: pre-wrap;">
            {{ lesson.content | linebreaksbr }}
          </div>
        </td>
        <td>
          {% if lesson in student.completed_lesson.all %}
            <span class="badge bg-success">Completed</span>
          {% else %}
            <span class="badge bg-secondary">Incomplete</span>
          {% endif %}
        </td>
        <td>
          <a href="{% url 'lesson_edit' lesson.id %}" class="btn btn-sm btn-outline-warning my-1">
            <i class="fas fa-edit"></i> Edit
          </a>

          <form action="{% url 'lesson_delete' lesson.id %}" method="POST" class="d-inline">
            {% csrf_token %}
            <button type="submit" class="btn btn-outline-danger btn-sm my-1">
              <i class="fas fa-trash-alt"></i> Delete
            </button>
          </form>
        </td>
      </tr>
      </tbody>
    </table>
  </div>

```

```

    {% if lesson not in student.completed_lesson.all %}

    <!-- Show Complete Button -->

    <form action="{% url 'complete_lesson' lesson.id %}" method="POST" class="d-inline">

        {% csrf_token %}

        <button type="submit" class="btn btn-outline-success btn-sm my-1">

            <i class="fas fa-check-circle"></i> Complete

        </button>

    </form>

    {% comment %} {% else %}

    <!-- Show Incomplete Button -->

    <form action="{% url 'uncomplete_lesson' lesson.id %}" method="POST" class="d-inline">

        {% csrf_token %}

        <button type="submit" class="btn btn-outline-secondary btn-sm my-1">

            <i class="fas fa-times-circle"></i> Incomplete

        </button>

    </form> {% endcomment %}

    {% endif %}

</td>

</tr>

{% empty %}

<tr>

    <td colspan="4" class="text-center text-danger fw-bold">No lessons available for this course.</td>

</tr>

{% endfor %}

</tbody>

</table>

</div>

<!-- Back Button -->

<div class="d-flex justify-content-center mt-4">

    <a href="{% url 'course_list' %}" class="btn btn-outline-primary">

        <i class="fas fa-arrow-left"></i> Back to Courses

    </a>

</div>

</div>

{% endblock content %}

```

Lab-8

Updated views.py using CourseListView

```
class CourseListView(LoginRequiredMixin, ListView):
    model = Course
    template_name = "Course_app/course_list.html"
    context_object_name = "courses"
    login_url = 'login_user'
```

Updated views.py using CourseDetailView

```
class CourseDetailView(LoginRequiredMixin, View):
    login_url = 'login_user' # Redirect to login if not authenticated

    def get(self, request, id):
        course = get_object_or_404(Course, id=id)
        lessons = course.lessons.all()

        # Get the logged-in student (assuming one-to-one relation with user)
        try:
            student = request.user.student
        except Student.DoesNotExist:
            student = None

        total_lessons = lessons.count()
        completed_lessons = (
            student.completed_lesson.filter(course=course).count() if student else 0
        )
        percent_complete = int((completed_lessons / total_lessons) * 100) if total_lessons > 0 else 0

        context = {
            'course': course,
            'lessons': lessons,
            'student': student,
            'total_lessons': total_lessons,
            'completed_lessons': completed_lessons,
            'percent_complete': percent_complete,
        }

        return render(request, "Course_app/Course_details.html", context)
```

Updated views.py using CourseCreateView

```
class CourseCreateView(LoginRequiredMixin, CreateView):
    model = Course
    form_class = CourseForm
    template_name = html_path_for_input_and_update
    login_url = 'login_user'
    success_url = reverse_lazy('course_list')

    def form_valid(self, form):
        response = super().form_valid(form)
        messages.success(self.request, "Course added successfully!")
        return response

    def get_context_data(self, **kwargs):
        context = super().get_context_data(**kwargs)
        context['check'] = 2 # Passing 'check' like in your FBV
        return context
```

Updated urls.py to include CBVs.

```
# For CBV
from .views import CourseListView, CourseDetailView, CourseCreateView

urlpatterns = [
    # path("", views.course_list, name='course_list'),
    path("", CourseListView.as_view(), name='course_list'),

    # path('details/<int:id>/', views.Course_Details, name='coursedetails'),
    path('details/<int:id>/', CourseDetailView.as_view(), name='coursedetails'),

    # path('create_course/', views.create_course, name='create_course'),
    path('create_course/', CourseCreateView.as_view(), name='create_course'),
]
```

Templates: course_list.html

```
{% extends 'Course_app/base.html' %}

{% block title %}Home Page{% endblock title %}

{% block content %}
```

```

<div class="container">

    <!-- Page Heading -->

    <div class="text-center mb-4">

        <h1 class="fw-bold text-primary">Welcome to <span class="text-success">Edu Learn</span></h1>

        <p class="text-muted">Manage courses, lessons, and student enrollments with ease.</p>

    </div>

    <!-- Action Buttons -->

    <div class="d-flex justify-content-between mb-4">

        <a href="{% url 'create_course' %}" class="btn btn-outline-primary fw-bold">
            <i class="fas fa-book"></i> Add New Course
        </a>

        <a href="{% url 'Enroll_student' %}" class="btn btn-outline-warning fw-bold">
            <i class="fas fa-user-plus"></i> Enroll Student
        </a>

    </div>

    <!-- Courses Table -->

    <div class="table-responsive">

        <table class="table table-bordered table-hover text-center align-middle">

            <thead class="table-dark">

                <tr>

                    <th>ID</th>

                    <th>Title</th>

                    <th>Description</th>

                    <th>Duration</th>

                    <th>Thumbnail</th>

                    <th>Actions</th>

                    <th>Manage Lesson</th>

                </tr>

            </thead>

            <tbody>

                {% for course in courses %}

                <tr>

                    <td>{{ forloop.counter }}</td>

                    <td class="fw-bold text-primary">{{ course.title }}</td>

                    <td class="text-muted">{{ course.descriptions | truncatewords:10 }}</td>

                    <td><span class="badge bg-success">{{ course.durations }} hrs</span></td>

                    <td>

                        {% if course.thumbnail %}


```



```

        
    {% else %}
        <span class="text-muted">No Image</span>
    {% endif %}
</td>
<td>
    <a href="{% url 'course_edit' course.id %}" class="btn btn-sm btn-outline-warning my-1">
        <i class="fas fa-edit"></i> Edit
    </a>
    <form action="{% url 'course_delete' course.id %}" class="d-inline">
        {% csrf_token %}
        <button type="submit" class="btn btn-outline-danger btn-sm my-1">
            <i class="fas fa-trash-alt"></i> Delete
        </button>
    </form>
</td>
<td>
    <a href="{% url 'create_lesson' %}" class="btn btn-outline-info btn-sm mb-1">
        <i class="fas fa-plus"></i> Add Lesson
    </a>
    <a href="{% url 'coursedetails' course.id %}" class="btn btn-outline-secondary btn-sm mb-1">
        <i class="fas fa-eye"></i> View Lessons
    </a>
    <a href="{% url 'individual_course_enrolled_student' course.id %}" class="btn btn-outline-primary
btn-sm mt-1">
        <i class="fas fa-users"></i> View Students
    </a>
</td>
</tr>
{% empty %}
<tr>
    <td colspan="7" class="text-center text-danger fw-bold">No courses available.</td>
</tr>
{% endfor %}
</tbody>
</table>
</div>

<!-- Back Button -->
<div class="d-flex justify-content-center mt-4">

```

```

<a href="{% url 'course_list' %}" class="btn btn-outline-success">
    <i class="fas fa-arrow-left"></i> Back to Dashboard
</a>
</div>
</div>
{% endblock content %}

```

Templates: course_detail.html,

```

{% extends 'Course_app/base.html' %}

{% block title %}Course Details{% endblock title %}

{% block content %}
<div class="container">
    <!-- Course Info -->
    <div class="card shadow-lg p-4 mb-4">
        <h2 class="text-primary fw-bold">{{ course.title }}</h2>
        <p class="text-muted mb-1">
            <strong>Duration:</strong>
            <span class="badge bg-success">{{ course.durations }} hours</span>
        </p>
        <p><strong>Description:</strong> {{ course.descriptions }}</p>
    </div>

    <!-- Lesson Progress -->
    <div class="mb-4">
        <h5>Progress: {{ completed_lessons }}/{{ total_lessons }} lessons completed</h5>
        <div class="progress">
            <div class="progress-bar bg-info" role="progressbar"
                style="width: {{ percent_complete }}%"
                aria-valuenow="{{ completed_lessons }}" aria-valuemin="0" aria-valuemax="{{ total_lessons }}">
                {{ percent_complete }}%
            </div>
        </div>
    </div>

    <!-- Add Lesson Button -->
    <div class="d-flex justify-content-between align-items-center my-3">

```

```

<h4 class="fw-bold text-secondary">Lessons</h4>

<a href="{% url 'create_lesson' %}" class="btn btn-success">
    <i class="fas fa-plus"></i> Add New Lesson
</a>
</div>

<!-- Lessons Table -->
<div class="table-responsive">
    <table class="table table-bordered table-hover text-center align-middle">
        <thead class="table-dark">
            <tr>
                <th>Lesson Title</th>
                <th>Content</th>
                <th>Status</th>
                <th>Actions</th>
            </tr>
        </thead>
        <tbody>
            {% for lesson in lessons %}
            <tr>
                <td class="fw-bold text-primary">{{ lesson.title }}</td>
                <td class="text-start">
                    <div class="border rounded p-2 bg-light" style="white-space: pre-wrap;">
                        {{ lesson.content | linebreaksbr }}
                    </div>
                </td>
                <td>
                    {% if lesson in student.completed_lesson.all %}
                        <span class="badge bg-success">Completed</span>
                    {% else %}
                        <span class="badge bg-secondary">Incomplete</span>
                    {% endif %}
                </td>
                <td>
                    <a href="{% url 'lesson_edit' lesson.id %}" class="btn btn-sm btn-outline-warning my-1">
                        <i class="fas fa-edit"></i> Edit
                    </a>

                    <form action="{% url 'lesson_delete' lesson.id %}" method="POST" class="d-inline">
                        {% csrf_token %}

```

```

        <button type="submit" class="btn btn-outline-danger btn-sm my-1">
            <i class="fas fa-trash-alt"></i> Delete
        </button>
    </form>

    {% if lesson not in student.completed_lesson.all %}
        <!-- Show Complete Button -->
        <form action="{% url 'complete_lesson' lesson.id %}" method="POST" class="d-inline">
            {% csrf_token %}
            <button type="submit" class="btn btn-outline-success btn-sm my-1">
                <i class="fas fa-check-circle"></i> Complete
            </button>
        </form>
    {% comment %} {% else %}
        <!-- Show Incomplete Button -->
        <form action="{% url 'uncomplete_lesson' lesson.id %}" method="POST" class="d-inline">
            {% csrf_token %}
            <button type="submit" class="btn btn-outline-secondary btn-sm my-1">
                <i class="fas fa-times-circle"></i> Incomplete
            </button>
        </form> {% endcomment %}
    {% endif %}
</td>
</tr>
{% empty %}
<tr>
    <td colspan="4" class="text-center text-danger fw-bold">No lessons available for this course.</td>
</tr>
{% endfor %}
</tbody>
</table>
</div>

<!-- Back Button -->
<div class="d-flex justify-content-center mt-4">
    <a href="{% url 'course_list' %}" class="btn btn-outline-primary">
        <i class="fas fa-arrow-left"></i> Back to Courses
    </a>
</div>
</div>

```

```
{% endblock content %}
```

Templates:course_form.html.

```
{% extends 'Course_app/base.html' %}

{% block title %}

    {% if check == 1 %} Update Course Details
    {% elif check == 2 %} Add New Course
    {% elif check == 3 %} Add New Lesson
    {% elif check == 4 %} Update Lesson Details
    {% elif check == 5 %} Enroll Student
    {% elif check == 6 %} Update Student
    {% endif %}

{% endblock title %}

{% block content %}
<div class="container mt-5">
    <div class="card shadow-lg border-0">
        <div class="card-body p-4">
            <h2 class="text-center text-primary fw-bold">
                {% if check == 1 %} Update Course Details
                {% elif check == 2 %} Add New Course
                {% elif check == 3 %} Add New Lesson
                {% elif check == 4 %} Update Lesson Details
                {% elif check == 5 %} Enroll Student
                {% elif check == 6 %} Update Student
                {% endif %}
            </h2>

            <!-- Display form errors -->
            {% if form.errors %}
                <div class="alert alert-danger">
                    <strong>Please correct the errors below:</strong>
                    <ul>
                        {% for field, errors in form.errors.items %}
                            {% for error in errors %}
                                <li>{{ field|title }}: {{ error }}</li>
                            {% endfor %}
                        {% endfor %}
                    </ul>
                </div>
            {% endif %}
        </div>
    </div>
</div>
{% endblock content %}
```

```

        {% endfor %}
    {% endfor %}
</ul>
</div>
{% endif %}

<form method="POST" enctype="multipart/form-data" class="mt-3">
    {% csrf_token %}
    <div class="mb-3">
        {{ form.as_p }}
    </div>
    <div class="d-flex justify-content-between">
        <button type="submit" class="btn btn-primary fw-bold">
            <i class="fas fa-check"></i> Submit
        </button>
        <a href="{% url 'course_list' %}" class="btn btn-secondary">
            <i class="fas fa-arrow-left"></i> Back to Home
        </a>
    </div>
</form>
</div>
</div>
{% endblock content %}

```

Course list page.

Home Page Password Reset OTP - EduLearn Select course to change | D... chemistry - Google Search

127.0.0.1:8000

YouTube Complete Competi... Complete Competi... CP DTC DLD DBMS Language Job site VIVA QUESTION 3-1 Project CV Interview_question Icon All Bookmarks

Edu Learn Home About Services Courses Students Course_wise_Student abdullah

Welcome to Edu Learn

Manage courses, lessons, and student enrollments with ease.

[Add New Course](#) [Enroll Student](#)

ID	Title	Description	Duration	Thumbnail	Actions	Manage Lesson
1	C++	C++ is a popular programming language. C++ is used to ...	60 hrs		Edit Delete	+ Add Lesson View Lessons View Students
2	Python	Python is a popular programming language. It was created by ...	40 hrs		Edit Delete	+ Add Lesson View Lessons View Students

Course detail page.

Course Details

127.0.0.1:8000/details/2/

Edu Learn

Home About Services Courses Students Course_wise_Student

abdullah

C++

Duration: 60 hours

Description: C++ is a popular programming language. C++ is used to create computer programmes and is one of the most used languages in game development. C++ was developed as an extension of C, and both languages have almost the same syntax.

Progress: 3/4 lessons completed

75%

Lessons

+ Add New Lesson

Lesson Title	Content	Status	Actions
What is C++?	C++ is a cross-platform language that can be used to create high-performance applications. C++ was developed by Bjarne Stroustrup, as an extension to the C language. C++ gives programmers a high level of control over system resources and memory. The language was updated 5 major times in 2011, 2014, 2017, 2020, and 2023 to C++11, C++14, C++17, C++20, and C++23.	Completed	EditDelete
	C++ is one of the world's most popular programming languages. C++ can be found in today's operating systems, Graphical User Interfaces, and embedded systems. C++ is an object-oriented programming language which gives a clear structure to programs and allows code		

Course creation form

Add New Course

Title:

Descriptions:

Durations:

Durations in hours

Thumbnail: No file chosen